# MOBIN ROZATI

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Backend Software Engineer focused on Python, Django, FastAPI, scalable backend architecture, and AI-enabled systems. Experience spans event-driven backend design, database optimization, LLM/MCP integrations, and applied machine learning/computer vision using Keras and OpenCV.

## WORK EXPERIENCE

**Backend Software Engineer | Cryptocurrency Trading Simulation Platform** **Feb 2026 – Jul 2026**

- Led the architectural refactoring of a Django-based cryptocurrency trading engine, addressing singleton execution, database hot paths, Redis coupling, and concurrency bottlenecks.
- Designed a deterministic, event-driven architecture with account-based execution sharding, durable event processing, idempotency, replay, and crash recovery.
- Developed a behavior-preserving migration strategy and compatibility tests for order execution, positions, margin, PnL, liquidation, drawdown rules, and real-time market-data processing.

**AI Software Engineering Intern | FAVA** **Jul 2025 – Sep 2025**

- Spearheaded the design and implementation of a FastMCP Server Integration project during an intensive 3-month summer internship.
- Connected an advanced MCP server with Large Language Models (LLMs), facilitating the seamless sharing of tools like RAG pipelines and external APIs.
- Enhanced the model's contextual retrieval capabilities, laying the foundation for scalable, AI-driven backend services.

**Backend Software Engineer | Hozhan Company** **Jun 2024 – Mar 2025**

- Engineered a secure, high-performance B2B backend architecture, resolving critical performance bottlenecks while ensuring strict backward compatibility with older legacy client technologies.
- Designed and optimized relational databases using PostgreSQL and MySQL, ensuring highly efficient data retrieval and synchronization across distributed services.
- Collaborated with cross-functional teams to integrate APIs and deploy containerized applications using Docker and Linux environments.

## EDUCATION

**B.Sc. Computer Engineering | K.N. Toosi University of Technology (KNTU)** **Sept 2022 – Present**

- Research Focus: Deep Reinforcement Learning (DRL) and Digital Twin (DT)-enabled Vehicular Edge Computing.

## TECHNICAL SKILLS

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|---|---|
| **Languages** | Java, Python, Go (Golang), SQL, C, English (Fluent) |
| **Backend Web** | FastAPI, Django, Django REST Framework, RESTful APIs, Celery, WebSockets, ASGI |
| **Databases & Search** | PostgreSQL, MySQL, Redis, MongoDB, Oracle Database |
| **Architecture** | OOP, Domain-Driven Design (DDD), Clean Architecture, SOLID, JWT, Microservices, Multithreading, Event-Driven Architecture, Transactional Outbox |
| **AI & Data** | Agentic AI, LLM Integration, RAG, LangChain, LangGraph, Web Crawling (BeautifulSoup, selenium), Prompt Engineering |
| **Computer Vision & ML** | PyTorch, OpenCV, CNNs, Image Processing, Image Restoration, Edge/Corner Detection, Homography & Perspective Geometry, SIFT/Feature Matching |
| **Testing & DevOps** | Unit Testing, Integration Testing, Compatibility Testing, Git, Docker, Linux (Fedora, Ubuntu), Postman |

## PROJECTS & RESEARCH

### URL Shortener – Backend Service

Go (Golang), REST APIs, PostgreSQL, GORM, Docker, Concurrency

- Developed a Go HTTP service for URL shortening, redirects, and link metadata, using a layered architecture with dependency injection and interchangeable in-memory and PostgreSQL storage.
- Implemented concurrency-safe storage, collision-resistant short-code generation, idempotent URL creation, and database uniqueness constraints to handle concurrent requests reliably.
- Built unit, integration, and race-condition tests, achieving 92.9% statement coverage with PostgreSQL integration tests in the documented verification environment.

### Document Scanning & Enhancement

Python, PyTorch, OpenCV, CNNs, Homography, Computer Vision

- Built an end-to-end document-scanning pipeline combining four-corner heatmap localization, perspective rectification, and residual encoder-decoder enhancement.
- Designed a paired synthetic-data pipeline with perspective transforms, backgrounds, illumination/shadow variation, blur, noise, JPEG degradation, and source-level train/validation/test separation.
- Achieved 29.35 dB PSNR / 0.934 canonical SSIM for synthetic enhancement and 0.53 px mean heatmap corner error with 99.5% all-corners success; evaluated real-world OCR/domain-gap failure modes and hardened the project with 173 automated tests.

### High-Performance CSV Data Pipeline

Java, MySQL, OOP

- Designed and implemented a memory-efficient backend utility engineered to stream and parse massive datasets concurrently without triggering out-of-memory errors.
- Developed a configuration engine enabling dynamic, user-defined schema rendering and conditional runtime relational-table compilation.
- Built a dual-stream transactional validation pipeline to isolate semantic parsing errors and log corrupt payloads directly into a structured error matrix without interrupting execution flow.

### AI-Powered Web Application (Mini GPT)

FastAPI, Hugging Face, Python

- Developed a seamless real-time communication platform integrating advanced open-source Language Models via Hugging Face.
- Implemented non-blocking REST APIs with FastAPI to handle concurrent user requests efficiently.

## CERTIFICATIONS

- AI Engineer for Developers Associate (AIEDA0010738576542) — DataCamp | datacamp.com/certificate/AIEDA0010738576542
- Model Context Protocol (MCP) Course Certificate (Aug 2025) — Hugging Face | Verify Certificate